PHY 102 Course Outline

1. Introduction to Semiconductors

- Definition of semiconductors
- Intrinsic and extrinsic semiconductors
- Doping
- p-type and n-type materials
- p-n junctions

2. Electrostatics

- Electric charge
- Electric field
- Electric field strength/intensity
- Coulomb's Law
- Electric field lines

3. Capacitors and Dielectrics

- Capacitance
- Energy stored in a capacitor
- Capacitors in series and parallel
- Dielectrics
- Drift velocity

4. Ohm's Law and Resistance

- Ohm's Law (V = IR)
- Resistors in series
- Resistors in parallel

5. Current Electricity

- Electric current
- Electromotive force (EMF)
- Internal resistance
- Cells in series and parallel

6. Alternating Current (AC)

- AC voltage and current
- Frequency and angular frequency
- RMS values

7. Components in AC Circuits

- Resistors (R)
- Capacitors (C)
- Inductors (L)

8. Special AC Circuits

- RC Circuit
- RL Circuit
- RLC Circuit
- Resonance